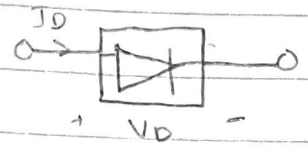
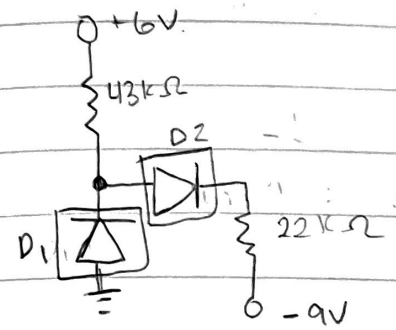
Homework # 5

1) Diode - Q point Analysis

a) ideal diode :





Assume D_1 to be off, D_2 to be on

↳ D_1 is reverse-biased : $I_{D_1} = 0$

$\therefore R_{total} = R_1 + R_2 = 43\Omega + 22k\Omega = 65k\Omega$

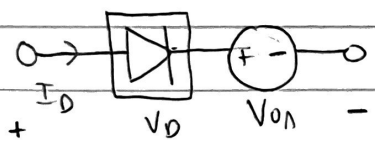
$$I_{D2} = \frac{V}{R_{total}} = \frac{6-(-9V)}{65k\Omega} = \sim 230\mu A$$

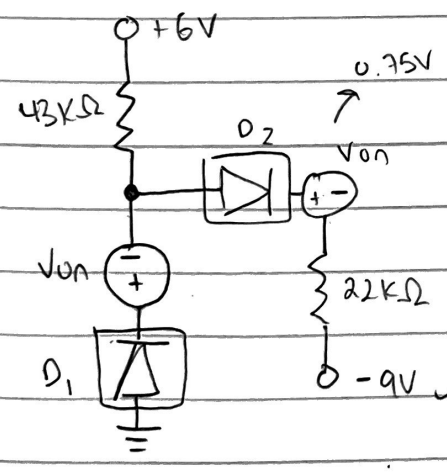
& $V_{D2} = 0$

$$V_{D1} = 6V - (43k\Omega \times I_{D2}) = \sim -3.92V$$

Q-point : $D_1(0, -3.92V)$ $D_2(230\mu A, 0V)$

b) CVD : Series connection of ideal diode and voltage source





↑ $V_{on} = 0.75V$

Same assumption as above : D_1 off & D_2 on

↳ D_1 is reverse-biased : $I_D = 0A$

$R_{total} = 65k\Omega$

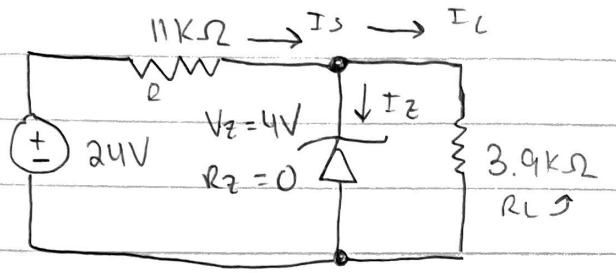
$$I_{D2} = \frac{V}{R_{total}} = \frac{6V - 0.75 - (-9V)}{65k\Omega}$$

$\therefore I_{D2} = 219\mu A$ & $V_{D2} = V_{on}$ for $I_D > 0$

$$V_{D1} = 6V - (43k\Omega \times I_{D2}) = \sim -3.43V$$

Q-point : $D_1(0A, -3.43V)$, $D_2(219\mu A, 0.75V)$

2) Zener Diode



Zener diode → Voltage Source (V_z) + Resistor (R_z)

$I_s = I_z + I_L$; $I_s = V_s - V_z / R = (24V - 4V) / 11K\Omega = 1.82mA$

$I_L = V_z / R_L = (4V) / 3.9K\Omega = 1.026mA$

↳ $\therefore I_z = \sim 0.792mA$

↳ Q-point : $(0.792mA, \overset{V_z}{4V})$

3) Half-wave Rectifier : 3.3V, 30A DC Power supply, ripple → 2.5% ∠ (60 hz) w/ $V_{on} = 1V$

a) Size of filter capacitor C : Ripple voltage → $V_r = I_{dc} \frac{T}{C}$

↳ $T = \frac{1}{f} \Rightarrow T = \frac{1}{60hz} = 0.0167s$; $\therefore C = I_{dc} \frac{T}{V_r} \Rightarrow C = (30A)\left(\frac{0.0167s}{0.0825V}\right)$

↳ $V_r = 0.025 \times 3.3V = 0.0825V$

↳ $C = 6.06F$

b) PIV rating : $PIV = 2V_p - V_{on} \cong 2V_p \cong 2(V_{dc} + V_{on})$

$V_{dc} = -(V_p - V_{on}) \rightarrow V_p = V_{dc} + V_{on}$

↳ 1 b/c large currents

$\therefore PIV \cong 2(3.3V + 1V) = 8.6V$

c) rms Value : $V_{rms} = \frac{V_p}{\sqrt{2}} \Rightarrow V_{rms} = \frac{(3.3V + 1V)}{\sqrt{2}} = 3.04V$

d) peak diode current : $\Delta T = \omega^{-1}\sqrt{\frac{2T}{RC}\left(\frac{V_p - V_{on}}{V_p}\right)}$; $I_p = I_{dc}\left(\frac{2T}{\Delta T}\right)$

↳ $\Delta T = \frac{1}{2\pi(60hz)}\left(\sqrt{\frac{2(1/60hz)}{0.11\Omega \times 6.06F}}\right)\left(\frac{4.3V - 1V}{4.3V}\right)$ w/ $V_{on} = 1V$

$R = \frac{3.3V}{30A} \approx 0.11\Omega$

$\therefore \Delta T = 0.520ms$ } $I_p = I_{dc}\left(\frac{2T}{\Delta T}\right) = (30A)\left(\frac{2 \times 1/60hz}{0.520ms}\right)$

$\therefore I_p \cong 1923.1\ A$

e) Surge current @ $t = 0^+$: $I_{sc} = \omega C V_p$

↳ $I_{sc} = (2\pi 60hz)(6.06F \times 4.3V) = 9823.6A$

4) Full-wave Rectifier Circuit : $V_{on} = 0.7V$; $R = 3\Omega$, $C = 22mF$
$rms\ (V_s) = 10V$, $f = 400hz$

DC output voltage : $V_{dc} = (V_p - V_{on}) \rightarrow V_{dc} = 10\sqrt{2}\ V - 0.7V = 13.44V$

↳ $V_{rms} = \frac{V_p}{\sqrt{2}} \rightarrow V_p = \sqrt{2}V_{rms} = 10\sqrt{2}$

Ripple voltage : $V_r = \frac{(V_p - V_{on})}{R} \frac{T}{2C} \rightarrow V_r = \left(\frac{13.44V}{3\Omega}\right)\left(\frac{1/400\ s}{2 \times 22mF}\right)$

↳ $V_r = (4.48\ V/\Omega) \times (0.057\ S/F) = 0.254V$

Peak Diode Current : $I_p = I_{dc}\left(\frac{2T}{2\Delta T}\right) \rightarrow I_p = \left(\frac{V_p - V_{on}}{R}\right)\left(\frac{2T}{2\Delta T}\right)$

↳ $I_p = (4.48\ V/\Omega) \times \left(\frac{2/400\ s}{0.504ms}\right) = 150A$

$\Delta T \cong \frac{1}{\omega}\sqrt{\frac{2T}{2RC}\left(\frac{V_p - V_{on}}{V_p}\right)} = \frac{1}{2\pi 400hz}\sqrt{\frac{\cancel{2}\cdot(1/400)}{\cancel{2}\ 3 \times 22mF}\left(\frac{13.44V}{10\sqrt{2}\ V}\right)}$

$\Delta T \cong 0.504ms$